Vytalix
Technology for Life

STRATEGY & OPERATIONS

VYTALIX FINANCIAL MODEL (36 months, three scenarios)

Vytalix company-build documentation · reference
08_FINANCIAL_MODEL

Status labels apply throughout: KNOWN, VERIFIED, ASSUMPTION, ESTIMATE, TO VALIDATE, PROPOSED. Nothing in this document claims traction, approvals, partners or revenue that do not exist. Governing file: FACTS_BASE.

PREPARED FOR	DATE	VERSION
The Founder, Vytalix	8 September 2026	1.0 · Confidential draft

Contents

Owner: CFO / Fundraising Director · **Status:** SCENARIO MODEL v1.1 (built 7 September 2026, revised 8 September 2026) · **Governing file:** `/FACTS_BASE.md` · **Workbook:** `/finance/Vytalix_Financial_Model_3yr.xlsx` (generated by `/finance/build_model.py`) · **Pricing source:** `/docs/01_BUSINESS_MODEL.md`

> **Truth standard.** Vytalix has no revenue, customers, contracts, grants, employees or product in production (FACTS_BASE §1, A3, A8). **Every number in this document is a scenario output of stated assumptions, not a forecast and not an actual.** GBP throughout. Model month 1 = January 2027 (assumes incorporation in Q4 2026). Tax and regulatory figures are labelled TO VALIDATE.

Executive view

1. **Three scenarios, one engine.** Conservative / Base / High-growth are built from ~100 explicit assumptions in one formula-driven workbook (scenario selector `Assumptions!C4`), mirrored line-for-line in Python so a "Static summary" sheet shows all three without recalculation. Base-case revenue is **£166k (FY2027) → £488k (FY2028) → £820k (FY2029)**; EBITDA **−£56k → −£190k → −£787k** as the seed is spent on scaling MaternaLink (in development).
2. **Services are the cash, MaternaLink is the equity story — the numbers confirm FACTS_BASE A6.** In Base, Advisory + Consulting are 59% of three-year revenue (£867k of £1.47m) and are gross-margin positive from month 2; MaternaLink is 26% (£387k) and does not earn a pound until month 13 at the earliest, or month 18 in Conservative.
3. **Cash requirement.** Peak cumulative cash need before any external money is **£229k (Conservative), £1.06m (Base), £145k (High-growth)** over 36 months. Base is funded by a £400k pre-seed (month 4) and a £2.0m seed (month 20); it closes month 36 with £1.42m and ~23 months' runway. Conservative survives 36 months on a £250k pre-seed only by not hiring a second technical salary, and ends with £21k (~2 months' runway) — the honest "services-only" outcome.
4. **Break-even is not reached within 36 months in Conservative or Base.** The workbook's "first month with positive EBITDA" (months 2–4) is a pre-salary blip; the honest metric, *sustained* positive EBITDA to month 36, is reached only in High-growth (month 31). Base is a deliberate investment case: cumulative EBITDA of −£1.03m buys a product, evidence and a team of 8.
5. **What actually moves value:** retainers (±£91k three-year EBITDA per retainer held in Y2–Y3), consulting project value (±£72k–£144k), and — once the company is scaled for it — MaternaLink go-live timing (a 12-month slip in High-growth turns three-year EBITDA from +£114k to −£661k and raises peak cash need from £145k to £787k). Founder day rate barely matters (±£5k) because founder capacity is fully consumed by retainers and projects — capacity, not price, is the constraint.

1. What the model is, and is not

Item	Position
Horizon	36 months, January 2027 – December 2029 (FY = calendar year). Month 1 assumes the company is incorporated, the name cleared and the founder available 4 days/week (208 days/year).
Entity	One UK company, GBP reporting (`02_GROUP_STRUCTURE.md` Option A). No subsidiary modelled.
Structure	Assumptions → Revenue build → Cost of sales → Headcount → Opex → P&L → Cash flow → Funding requirement → Dashboard; Sensitivity and Static summary are Python-computed values. README sheet documents conventions and the 8 September change log.

Item	Position
Revenue logic	Advisory = founder billable days × day rate (after retainer and project days) + retainers × fee + senior consultant days (once hired). Consulting = projects/year × average value. E-learning = new learners × ARPU (compounding monthly from launch) + corporate licences. MaternaLink = paid pilots × annual value (from first-pilot month) + per-woman licences (UK and Africa) from go-live month.
Cost logic	Cost of sales per stream; employment = salaries + employer NI 15% above £5,000 (KNOWN, TO VALIDATE) + 3% auto-enrolment pension on qualifying earnings + Employment Allowance from 2+ employees (TO VALIDATE eligibility); ten opex categories by year; one-off legal/IP set-up (month 1) and e-learning content build (launch month).
Cash logic	Services and MaternaLink invoices collected with a one-month lag; e-learning prepaid; costs paid in month incurred; equity and grants received in the stated month.
Not modelled	VAT, corporation tax, depreciation, interest, R&D tax relief (upside, TO VALIDATE), working-capital beyond the collection lag, FX on Nigerian revenue (all GBP), bad debt on government receivables (a real risk — see §8).
Regulatory cost	Modelled inside "Compliance & regulatory" opex (ICO, Cyber Essentials, DTAC/DSPT, clinical safety) — a UK MDR/MHRA SaMD pathway for the v3 AI track is **not** modelled and would be a separate, grant-funded budget.
Excluded by policy	The Ekiti proposal's £300 per woman per year (FACTS_BASE A5) and its £4.5m/£9m/£36m phases. Africa licences are modelled at £3–£12 per woman per year; UK at £8–£40.

1.1 Revision log (8 September 2026 sanity check)

The script was repaired (a build-order fault: the P&L grant row and the Opening-cash row referenced rows not yet built; the builder now runs a two-pass layout). Six assumption groups were then adjusted because the first output was internally inconsistent, not merely optimistic. Each change is annotated in the workbook's Assumptions notes and README change log.

#	Was	Now	Why
1	Opening cash £10k/£20k/£30k; founder draw from month 4/2/1	£25k/£35k/£50k (incl. director's loan); draw from month 4/3/3	Cash was negative in months 2–5 of every scenario before the pre-seed landed. Founder draw now starts once month-2 invoices are collected (one-month lag).
2	Conservative hires Developer 1 in month 19	Not hired (contractors)	With a £250k pre-seed, cash went negative at month 31 (−£81k at month 36).
3	Base Y3 and High-growth Y2–Y3 opex at "lean" levels	Contractors, technology, marketing, compliance, travel and operations raised to scale-up levels	Base burned only ~£20k/month after a £2m seed and closed with £2.1m unspent; High-growth was EBITDA-positive from month 2 with £11.4m unspent at month 36 — money raised but never deployed is not a credible plan.
4	Series A £6m in month 32 (High-growth)	Removed from the 36-month model (inputs retained)	It was raised while cash-generative with £4m in the bank. No scenario needs a Series A for solvency inside 36 months; it is a month 30–42 growth-capital decision (`09_FUNDING_STRATEGY.md`).
5	Break-even = first month EBITDA > 0	Added "sustained" break-even (EBITDA positive from that month to month 36) to P&L, Funding requirement, Dashboard and Static summary	The first-positive month (2–4) is a pre-salary artefact and would mislead an investor.
6	Formula checker flagged 8 false positives	Fixed	`&Assumptions!D4` inside string concatenation was parsed as a sheet name.

Not adjusted, but flagged: in Base Y2–Y3 the founder is over-allocated (2 retainers × 2.5 days + 10 projects × 7 days ÷ 12 ≈ 10.8 days/month against 7.3 planned billable days). Either founder utilisation rises to ~60% in year 3 or associates deliver more of the project work (cost-of-sales % would rise). This is why founder day rate shows almost no sensitivity (§6).

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2. Assumptions per scenario

All values ESTIMATE / ASSUMPTION / PROPOSED unless marked KNOWN; full list with notes is on the Assumptions sheet (Conservative | Base | High-growth).

2.1 Services (cash layer)

Assumption	Conservative	Base	High-growth	Source / note
Founder available days/month	17.33	17.33	17.33	208 days/yr ÷ 12 (4 days/week)
Founder billable utilisation	30%	42%	55%	`01_BUSINESS_MODEL` §4.2 (42% ≈ 87 days/yr)
Founder day rate	£1,200	£1,500	£1,800	Range in `01_BUSINESS_MODEL` §4.1
Sales ramp (months to full run-rate)	4	3	2	Month 1 = £0 (incorporation, outreach)
Retainers active Y1 / Y2 / Y3 (average)	0.5 / 1 / 1	1 / 2 / 2	2 / 3 / 3	First retainer month 3 / 2 / 2
Retainer fee per month	£3,500	£4,000	£5,000	£2,500–£6,000 range
Founder days per retainer per month	2.5	2.5	2.5	
Consulting projects won Y1 / Y2 / Y3	2 / 4 / 5	5 / 8 / 10	8 / 14 / 18	Spread evenly after ramp
Average project value	£15,000	£20,000	£30,000	Catalogue prices £9.5k–£28k
Founder days per project	8	7	5	Associates deliver more in High
Consulting delivery cost (% of value)	30%	35%	45%	Associates and expenses
Senior consultant (billable) hire month	never	19	10	10 billable days/month at £900–£1,000
Direct cost per founder billable day	£50	£50	£50	Travel, tools

2.2 E-learning

Assumption	Conservative	Base	High-growth	Note
Launch month	12	9	7	Two founder-authored courses first
New learners per month at launch	8	15	30	
Monthly growth in new learners	3%	5%	7%	Compound from launch
ARPU per learner	£150	£200	£300	Courses £49–£299 UK; regional pricing lower
Corporate licences Y1 / Y2 / Y3	0 / 1 / 2	0 / 3 / 6	1 / 6 / 12	£2,500 / £5,000 / £9,000 per licence-year
Platform + payment fees	15%	15%	15%	of revenue

Assumption	Conservative	Base	High-growth	Note
Content build (one-off at launch)	£8,000	£12,000	£20,000	£8k–£15k per course

2.3 MaternaLink (in development) — equity layer

Assumption	Conservative	Base	High-growth	Note
First paid pilot month	18	13	10	TO VALIDATE — gated by IP, safety case, DPIA, partner site
Active paid pilots Y1 / Y2 / Y3	0 / 1 / 1	0 / 1 / 2	1 / 2 / 4	
Pilot value per pilot-year	£40,000	£75,000	£150,000	Recognised evenly while active
Pilot delivery cost (% of value)	50%	45%	40%	First pilots near loss-leaders
Per-woman licence go-live month	27	19	15	TO VALIDATE; `ml_delay` input tests slippage
UK women at go-live / added per month	200 / 50	500 / 100	1,500 / 300	
UK licence per woman per year	£8	£20	£40	≈ £40k–£120k per trust at 3,000–5,000 births
UK hosting/support per woman-year	£4	£4	£4	
Africa women at go-live / added per month	1,000 / 250	5,000 / 1,000	15,000 / 3,000	LGA-level, donor- or state-funded cohorts
Africa licence per woman per year	£3	£6	£12	Ekiti's £300 is UNVALIDATED and not used (A5)
Africa hosting/SMS/support per woman-year	£2.00	£2.50	£3.00	SMS pass-through excluded

2.4 People, overheads and funding

Assumption	Conservative	Base	High-growth	Note
Founder salary (start month)	£36k (m4)	£48k (m3)	£60k (m3)	Below market; opportunity cost £120k
Product lead / CTO (start, salary)	m13, £60k	m7, £65k	m4, £70k	
Developer 1 / 2 / 3	never / never / never	m10 / m22 / never	m6 / m12 / m24	£50k–£60k each
BD lead; Programme manager (Nigeria); Clinical lead; Content lead	all never	m16; m20; m25; never	m10; m14; m16; m18	Nigeria role modelled on UK on-costs (conservative simplification, TO VALIDATE)
Employer NI 15% above £5,000; pension 3%; Employment Allowance £10,500 from 2+ employees	KNOWN (2026/27) — TO VALIDATE against gov.uk			
Other opex, monthly Y1 → Y3 (total of ten categories)	~£3.2k → ~£15k	~£6.0k → ~£76k	~£8.8k → ~£135k	Contractors, technology, marketing, legal, compliance, insurance, software, travel, operations, accountancy

Assumption	Conservative	Base	High-growth	Note
Legal & IP set-up (month 1)	£6,000	£10,000	£15,000	Incorporation, SHA, IP assignment, SEIS/EIS AA, trade mark
Opening cash (founder + director's loan)	£25,000	£35,000	£50,000	TO VALIDATE with founder
Pre-seed equity (month)	£250k (m6)	£400k (m4)	£500k (m3)	PROPOSED; SEIS £250k company cap TO VALIDATE
Seed equity (month)	none	£2.0m (m20)	£3.0m (m14)	Only if `09_FUNDING_STRATEGY` milestones are TRUE
Series A	none in model	none in model	none in model	Month 30–42 decision; inputs retained for testing
Grants (month)	none	£75k (m14)	£150k (m9) + £250k (m21)	TO VALIDATE; competitive; none applied for
Collection lag	1 month	1 month	1 month	E-learning prepaid

3. Three-year annual P&L per scenario

Values from the workbook's Static summary sheet (Python-computed; identical logic to the formula sheets). Rounded to the nearest £1k in the text, exact in the tables.

3.1 Conservative — "services only, one hire, no seed"

£	FY2027	FY2028	FY2029	3-year
Advisory	46,569	42,466	42,000	131,034
Consulting	23,750	60,000	75,000	158,750
E-learning	1,200	20,041	30,010	51,251
MaternaLink (in development)	0	23,333	48,146	71,479
Total revenue	**71,519**	**145,840**	**195,156**	**412,514**
Cost of sales	(9,141)	(34,192)	(53,460)	(96,793)
Gross profit	**62,377**	**111,648**	**141,696**	**315,721**
Gross margin	87%	77%	73%	77%
Employment cost	(31,157)	(100,614)	(100,614)	(232,384)
Total operating expenses	**(84,037)**	**(188,814)**	**(282,414)**	**(555,264)**
EBITDA (before grants)	**(21,660)**	**(77,166)**	**(140,718)**	**(239,543)**
Grant income	0	0	0	0
Net result before tax	**(21,660)**	**(77,166)**	**(140,718)**	**(239,543)**
Net operating cash flow	(29,048)	(81,649)	(143,175)	(253,872)

£	FY2027	FY2028	FY2029	3-year
Closing cash (year end)	245,952	164,302	21,128	21,128
Headcount (year end)	1	2	2	2

Reading: a founder at 30% utilisation with one retainer, 2–5 projects a year and a product lead from month 13 cannot build MaternaLink; the £250k pre-seed is consumed by month 36 with a product lead, contractors and one late, small pilot. This is the scenario if the seed is not raised — and it says the company must choose by month 24 between a lifestyle consultancy and stopping product spend.

3.2 Base — "services fund the company, seed funds the product"

£	FY2027	FY2028	FY2029	3-year
Advisory	69,748	150,000	204,000	423,748
Consulting	83,333	160,000	200,000	443,333
E-learning	12,930	73,042	134,235	220,208
MaternaLink (in development)	0	105,000	282,000	387,000
Total revenue	**166,012**	**488,042**	**820,235**	**1,474,289**
Cost of sales	(33,339)	(114,581)	(208,485)	(356,406)
Gross profit	**132,672**	**373,461**	**611,750**	**1,117,883**
Gross margin	80%	77%	75%	76%
Employment cost	(94,785)	(290,117)	(485,583)	(870,484)
Total operating expenses	**(188,185)**	**(563,117)**	**(1,398,783)**	**(2,150,084)**
EBITDA (before grants)	**(55,512)**	**(189,656)**	**(787,033)**	**(1,032,201)**
Grant income	0	75,000	0	75,000
Net result before tax	**(55,512)**	**(114,656)**	**(787,033)**	**(957,201)**
Net operating cash flow	(70,639)	(217,780)	(804,616)	(1,093,035)
Closing cash (year end)	364,361	2,221,581	1,416,965	1,416,965
Headcount (year end)	3	7	8	8

Reading: FY2027 revenue of £166k is consistent with the £118k–£210k services range in `01_BUSINESS_MODEL` §4.3 plus a small e-learning contribution. FY2029 is the first year in which MaternaLink revenue (£282k: two pilots at £75k plus £132k of per-woman licences from ~22,000 women in Africa and ~2,200 in the UK) is material, and it is still only 34% of revenue. The FY2029 loss of £787k is the seed being spent as intended: a team of 8, £30k/month of implementation contractors and £15k/month of marketing.

3.3 High-growth — "everything lands early and the seed is reinvested"

£	FY2027	FY2028	FY2029	3-year
Advisory	161,567	300,000	300,000	761,567
Consulting	210,000	420,000	540,000	1,170,000

£	FY2027	FY2028	FY2029	3-year
E-learning	68,880	295,612	652,156	1,016,647
MaternaLink (in development)	37,500	680,000	1,584,000	2,301,500
Total revenue	**477,947**	**1,695,612**	**3,076,156**	**5,249,714**
Cost of sales	(123,181)	(438,592)	(794,423)	(1,356,196)
Gross profit	**354,766**	**1,257,020**	**2,281,732**	**3,893,518**
Gross margin	74%	74%	74%	74%
Employment cost	(190,632)	(545,608)	(655,693)	(1,391,933)
Total operating expenses	**(330,632)**	**(1,175,608)**	**(2,273,293)**	**(3,779,533)**
EBITDA (before grants)	**24,133**	**81,412**	**8,440**	**113,985**
Grant income	150,000	250,000	0	400,000
Net result before tax	**174,133**	**331,412**	**8,440**	**513,985**
Net operating cash flow	(30,524)	(4,931)	(74,560)	(110,015)
Closing cash (year end)	669,476	3,914,545	3,839,985	3,839,985
Headcount (year end)	6	10	10	10

Reading: this is an upside envelope, not a plan. It requires four paid pilots at £150k, 78,000 enrolled women in Africa at £12 and 7,800 in the UK at £40 by month 36, an e-learning business growing 7% a month for 30 months, and two grants totalling £400k — each individually possible, jointly improbable. Its purpose is to show what the equity story could look like and where the cost base would have to go (~£135k/month of non-people opex in FY2029). Even here, three-year EBITDA is only +£114k because growth is reinvested.

3.4 Cross-scenario summary

Metric	Conservative	Base	High-growth
3-year revenue	£412,514	£1,474,289	£5,249,714
Services share of 3-year revenue (Advisory + Consulting)	70%	59%	37%
MaternaLink share of 3-year revenue	17%	26%	44%
3-year EBITDA	(£239,543)	(£1,032,201)	£113,985
Peak cumulative cash need before external funding	£228,872 (m36)	£1,058,035 (m36)	£144,705 (m31)
Equity raised in model	£250,000	£2,400,000	£3,500,000
Grants in model	£0	£75,000	£400,000
Minimum closing cash after funding (month)	£4,726 (m5)	£6,099 (m3)	£12,750 (m2)
Closing cash, month 36	£21,128	£1,416,965	£3,839,985
Runway at month 36 (trailing 3-month burn)	1.9 months	23 months	cash-generative
First month EBITDA > 0 (blip)	3	4	2

Metric	Conservative	Base	High-growth
Sustained EBITDA break-even	not within 36 months	not within 36 months	month 31
Cumulative result positive	not within 36 months	not within 36 months	month 9 (grant-assisted)
Headcount, month 36	2	8	10

4. Cash requirement and funding

4.1 How much, and when

"Peak cash need" is the largest cumulative deficit of opening cash plus operating cash flow with **no** equity or grants — i.e. the total external money the scenario consumes over 36 months.

	Conservative	Base	High-growth
Peak cash need (month)	£229k (m36, still falling)	£1,058k (m36, still falling)	£145k (m31)
Cash need to month 12 alone	~£4k	~£36k	~£5k
Cash need to month 24 alone	~£86k	~£253k	~£64k
Funding in model	Pre-seed £250k (m6)	Pre-seed £400k (m4) + Seed £2.0m (m20) + grant £75k (m14)	Pre-seed £500k (m3) + Seed £3.0m (m14) + grants £400k
Lowest cash after funding	£4.7k (m5, the month before the pre-seed)	£6.1k (m3, the month before the pre-seed)	£12.8k (m2)
Funded-through check	Covers 36 months, ends with ~2 months' runway	Covers 36 months, ends with 23 months' runway	Covers 36 months, cash-generative

Three conclusions the CFO draws from this:

1. **The first 90 days are the tightest point in every scenario.** Minimum cash is £5k–£13k in months 2–5, before the pre-seed lands. The opening cash assumption (£25k–£50k of founder money) and the founder draw are the two numbers that decide whether the company is solvent in spring 2027. Re-running Base with the pre-seed slipping from month 4 to month 7 still leaves minimum cash positive (~£2.5k in month 4) because services receipts cover the pre-product cost base — the real rule is that **the product-lead hire (month 7 in Base) must not precede the money**.
2. **The pre-seed alone buys 36 months only in Conservative, and only by not building the product.** Base spends £253k by month 24 and £1.06m by month 36 — the seed is what funds MaternaLink's build, evidence and team. Without it, Base collapses into Conservative by month 20.
3. **The 20–30% buffer rule.** The rounds in the model are sized ~1.5–2× the modelled need to month 36 (Base: £2.4m raised vs £1.06m need) precisely because the need is at the *end* of the window and still growing at ~£65k/month; the buffer is 18–24 months of post-seed runway, which is what seed investors expect.

4.2 Rounds table (PROPOSED; every round conditional on 09_FUNDING_STRATEGY.md milestones)

Round	Conservative	Base	High-growth	Cumulative cash before the round (Base)
Pre-seed (SEIS/EIS angels)	£250k, month 6 (Jun 2027)	£400k, month 4 (Apr 2027)	£500k, month 3 (Mar 2027)	£6k
Seed	—	£2.0m, month 20 (Aug 2028)	£3.0m, month 14 (Feb 2028)	£306k
Series A	—	— (month 30–42 decision)	— (month 30–42 decision)	—
Grant 1	—	£75k, month 14	£150k, month 9	
Grant 2	—	—	£250k, month 21	

5. Break-even

Definition	Conservative	Base	High-growth
First month EBITDA > 0 (workbook row "EBITDA positive")	Month 3	Month 4	Month 2
Sustained EBITDA > 0 to month 36 (the honest metric)	**Not reached**	**Not reached**	**Month 31 (Jul 2029)**
Cumulative net result > 0	Not reached	Not reached	Month 9 (grant-assisted)
Services-only break-even (founder + tools, no product spend)	~5 founder billable days or 2 retainers per month (`01_BUSINESS_MODEL` §3) — achieved by month 3–4 in all scenarios		

Interpretation:

- The **services business breaks even almost immediately** — that is the point of it. Fixed cost is one founder draw and ~£3k–£6k/month of overheads.
- The **company does not break even** in Conservative or Base because it chooses to invest in MaternaLink. In Base, the monthly loss grows from ~£5k (FY2027) to ~£65k (FY2029). At month 36 MaternaLink earns ~£27k/month and is growing by only ~£0.7k/month (1,100 women added a month at £6–£20), so at that trajectory the ~£60k/month gap does not close for many years. **Break-even beyond month 36 therefore depends on pilot-to-licence conversion accelerating sharply (the High-growth trajectory) or on the cost base being held flat** — stated here as arithmetic, not a forecast.
- **What would bring Base break-even inside 36 months:** any two of (a) a third retainer held through Y2–Y3 (+£91k EBITDA), (b) average project value £30k instead of £20k (+£144k), (c) Africa licence at £12 rather than £6 (+£122k), (d) holding FY2029 non-people opex at FY2028 levels (+£560k, but that means not scaling). Only (d) gets close on its own, which is the real message: **break-even is a choice about scaling speed, not a property of the model.**

6. Sensitivity findings

All from the Sensitivity sheet (Python-computed). Base scenario unless stated; three-year totals.

6.1 One-at-a-time flex (Base: revenue £1,474k; EBITDA −£1,032k; peak need £1,058k)

Input	Low → High	Δ 3-yr revenue	Δ 3-yr EBITDA	Δ peak cash need
Retainers held in Y2/Y3	1 → 3	−£94k / +£96k	−£91k / +£93k	+£87k / −£89k
Average consulting project value	£15k → £30k	−£111k / +£222k	−£72k / +£144k	+£68k / −£136k
MaternaLink pilot value per year	£40k → £150k	−£105k / +£225k	−£58k / +£124k	+£52k / −£111k
Africa licence £/woman/yr	£3 → £12	−£61k / +£122k	−£61k / +£122k	+£55k / −£111k
Africa women added per month	250 → 3,000	−£57k / +£153k	−£33k / +£89k	+£27k / −£72k
E-learning ARPU	£150 → £300	−£44k / +£88k	−£37k / +£74k	+£37k / −£74k
UK licence £/woman/yr	£8 → £40	−£24k / +£41k	−£24k / +£41k	+£22k / −£37k
Founder utilisation	30% → 55%	−£26k / +£33k	−£25k / +£32k	+£25k / −£32k
Founder day rate	£1,200 → £1,800	−£5k / +£5k	−£5k / +£5k	+£5k / −£5k

6.2 Day rate × utilisation grid (Base, 3-year revenue)

	Utilisation 30%	42%	55%
£1,200/day	£1,449k	£1,469k	£1,495k
£1,500/day	£1,449k	£1,474k	£1,507k
£1,800/day	£1,449k	£1,479k	£1,519k

The grid is nearly flat because the founder's billable capacity is consumed by retainers and projects before any "advisory day" is sold (see §1.1). The lever is **capacity** — a billable senior consultant, associates, or fewer founder days per project — not price.

6.3 MaternaLink go-live delay

Delay	Base: 3-yr MaternaLink revenue	Base: 3-yr EBITDA	Base: peak need	High-growth: 3-yr EBITDA	High-growth: peak need
0 months	£387k	−£1,032k	£1,058k	+£114k	£145k
+3 months	£345k	−£1,059k	£1,083k	−£122k	£327k
+6 months	£309k	−£1,082k	£1,104k	−£330k	£495k
+12 months	£255k	−£1,116k	£1,134k	−£661k	£787k

In Base a 12-month slip costs £132k of revenue and £76k of peak need — survivable, because Base has not yet scaled for MaternaLink. In High-growth the same slip costs £775k of EBITDA and £642k of cash, because the cost base has been built ahead of the revenue. **Rule derived: do not scale the MaternaLink cost base (in-country team, marketing, hosting at volume) until go-live is contractually dated.** In Conservative the delay is almost irrelevant (£8k), which is its own warning: Conservative barely has a product story.

6.4 Other sensitivities worth knowing (computed by re-running compute with single changes)

- **Grants at zero in Base:** peak need (which by definition excludes all funding) is unchanged; closing cash at month 36 falls by exactly £75k. Grants are upside, never load-bearing — which is how they should be treated.
- **Collection lag 1 → 2 months (Base):** month-3 minimum cash falls from +£6k to about −£1k and peak cash need rises by £60k (the permanent receivables balance). Government and NHS receivables are frequently 60–90 days; this is a live risk for FY2028–29 MaternaLink invoices.
- **Employer NI at 13.8% instead of 15% (if the rate were to change):** ~£8k over three years in Base. Immaterial at this scale.

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7. Honest verdict on value drivers

1. **Retainers are the best money in the company.** One retainer held for two years is worth ~£91k of EBITDA — more than any single MaternaLink assumption inside 36 months — at ~63% margin and near-zero capex. The sales engine (`13_SALES_ENGINE.md`) should be measured on retainers first.
2. **Consulting project value beats project count.** Moving the average from £20k to £30k (i.e. selling C3/C8-type reviews rather than C7 sprints) adds £144k of EBITDA; winning two more projects a year adds less and consumes founder days the model has already over-allocated.
3. **MaternaLink is 26% of Base revenue and roughly 100% of the equity valuation.** Its three-year revenue (£387k) is smaller than Advisory alone; its cost (product lead, developers, BD lead, clinical lead, programme manager, contractors, technology, compliance ≈ £1.4m of the £2.15m three-year opex) is the reason the company loses £1.03m. An investor is not buying the FY2029 P&L; they are buying the milestones — IP ownership, safety case, design partner, paid pilot, evidence — that convert MaternaLink from a cost into a licensable asset. `09_FUNDING_STRATEGY.md` therefore prices each round on milestones, not multiples.
4. **The Africa price per woman is the single largest long-run unknown.** At £3 the Africa programme is a donor-funded evidence engine that never covers its fixed team; at £12 it is a business. Every comparable (mDoc/MSD for Mothers in Ekiti and Lagos, Jacaranda PROMPTS at under $1 per mother in Kenya — VERIFIED / ESTIMATE in `05_MARKET_STRATEGY.md`) points to the low end. The model's Base £6 is a hope, not a benchmark.
5. **E-learning is underrated inside the company and overrated outside it.** It is 15% of Base revenue at ~85% gross margin and needs no regulatory clearance — but the 5%/month compounding learner growth for 27 months is unproven, CAC is untested (target < £60), and the corporate-licence ladder has no buyer yet.
6. **The founder is the binding constraint in every scenario.** Services revenue, grant writing, MaternaLink product ownership, IP negotiation and fundraising all draw on the same 208 days. The model's over-allocation in Base Y2–Y3 (§1.1) is the quantitative expression of the risk stated in `01_BUSINESS_MODEL` §8 #1.
7. **What the model cannot tell you:** whether anyone will pay. Zero customers, zero pilots, zero letters of intent (FACTS_BASE §1). The first three signed retainers and the first design-partner letter will do more to de-risk this model than any sensitivity table.

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8. Assumptions to validate first (ranked by impact × uncertainty)

Rank	Assumption	Model value (Base)	Why it matters	How to validate (owner, by when)	Kill / pivot signal
1	Founder can hold 2 retainers and win 5 projects in year 1 at £1,500/day	Retainers 1 (Y1) → 2; 5 projects × £20k	59% of 3-year revenue; the whole cash layer	30 conversations, 6 proposals, 2 signed by day 90 (founder, month 3)	< 1 signed by month 3 → part-time, fractional role
2	MaternaLink IP can be brought into the company	Product spend from month 7	No IP = no product = no equity story; every product cost is wasted	Solicitor-led negotiation with David Agunede (founder + solicitor, month 2)	No agreement by month 4 → rebuild fresh or drop from equity story
3	Government / donor willingness to pay per woman in Nigeria	£6/woman/yr	±£122k EBITDA; decides whether Africa is a market or an evidence programme	Interviews with MSD for Mothers, state MoH digital leads, GFF country teams; mDoc's public scale (founder, month 9)	WTP < £3 → Africa is donor-funded only; drop from revenue model
4	A UK maternity unit will act as unpaid design partner, then paid pilot at £75k/yr	First paid pilot month 13	Pilot revenue £387k over 3 years; the reference every later sale needs	10 approaches via Health Innovation Networks and clinical advisors (founder + clinical advisor, month 9)	Zero interest → private/Africa-first
5	Pre-seed £400k closes by month 4	Cash minimum £6k in month 3	Solvency in spring 2027	SEIS/EIS advance assurance by month 2; 25 angel conversations by month 3	Not closed by month 6 → cut founder draw, defer product lead
6	E-learning learner growth 5%/month at CAC < £60	15 learners/month at launch, ARPU £200	£220k revenue; 85% margin	£1,000 paid + organic test around course 2 (founder, month 10)	CAC > £150 → B2B-only distribution
7	A grant of £75k is winnable by month 14	£75k, month 14	£75k of peak need; credibility	≥ 4 bids in 12 months (Innovate UK Smart, SBRI Healthcare, NIHR i4i Connect if an NHS partner exists, one foundation)	0/4 → remove from model
8	Employer NI 15%, threshold £5,000, Employment Allowance £10,500, pension 3%	KNOWN (2026/27)	Small (< £10k/yr) but appears in every payroll	Accountant to confirm against gov.uk (month 1)	—
9	SEIS eligibility (£250k company cap, £200k investor cap, < 3 years trading, gross assets < £350k) and EIS limits	Pre-seed structure	Angels expect relief; refusal cuts the angel pool	Advance assurance application (accountant, month 2–3)	Refusal → restructure or reduce angel expectations
10	One-month collection lag on NHS and government invoices	1 month	Month-3 cash and FY2028–29 working capital	Check payment terms in first pilot contract; assume 60 days for government	Lag > 2 months → invoice milestones in advance

9. Known limitations of the model

- Africa revenue is in GBP with no FX, no withholding tax, no in-country entity cost and no bad-debt provision; a Nigerian programme contract would need all four (TO VALIDATE).

- The Nigeria programme manager is costed on UK on-costs — conservative on cost, wrong on structure (likely a local contractor or partner).
- No VAT (services to UK businesses would carry VAT once registered; e-learning to consumers carries VAT on digital services), no corporation tax (loss-making; R&D relief unmodelled), no depreciation.
- Learner growth, women-added-per-month and pilot counts are smooth; reality is lumpy and contract-driven.
- No churn on retainers or subscriptions; no price escalation; no salary inflation.
- Founder capacity over-allocation in Base/High Y2–Y3 (§1.1) is flagged, not resolved.
- The workbook was rebuilt and re-opened with openpyxl (12 sheets, 6,827 formulas, 0 dangling references). A LibreOffice recalculation could not be run in the build environment (no Calc component); the `--verify` option in the script performs it where LibreOffice Calc is installed. Static summary values are Python-computed and are the figures used in this document.

Priorities / Risks / Next actions

Priorities

1. Treat the model as a **decision tool, not a forecast**: re-run monthly against actuals from month 2 (`operations/KPI_DASHBOARD.csv`) and re-label assumptions as they are validated or killed.
2. Sell retainers before anything else — the highest-value, lowest-risk line in every scenario.
3. Close the pre-seed by month 4 (Base) or cut the founder draw; month 3 is the solvency pinch-point in every scenario.
4. Resolve MaternaLink IP before month 7 — the month the first product salary starts in Base.
5. Do not scale the MaternaLink cost base until a go-live date is in a signed contract (§6.3).

Risks

- Founder bandwidth (over-allocated in Base Y2–Y3) — mitigate with a billable senior consultant from month 19 and associate-delivered projects.
- Africa willingness to pay materially below £6/woman/year — mitigate by treating Africa as a grant/donor-funded evidence market until validated.
- Pre-seed slips past month 6 — mitigate with founder bridge, deferred draw, deferred product lead.
- NHS/government receivables at 60–90 days — mitigate with milestone invoicing and advance payments in pilot contracts.
- Regulatory drift of MaternaLink into medical-device territory — unmodelled cost and a revenue freeze; mitigate via the intended-use statement and gates in `product/MATERNALINK_ROADMAP.md`.

Next actions (owner: CFO unless stated)

- Week 1: accountant confirms NI/pension/Employment Allowance rates and SEIS/EIS eligibility (items 8–9 in §8).
- Week 2: founder confirms opening cash and acceptable draw; update `opening_cash` and `founder_start`; re-run `python3 finance/build_model.py`.
- Month 1: open the actuals tracker (revenue, billable days, retainers, cash) mirroring the model rows; first variance review at month 3.

- Month 2: version the model as v1.2 once the IP route (assignment vs licence) is known — a licence with royalty changes MaternaLink cost of sales.
- Month 3: run the model on a machine with LibreOffice Calc (`python3 finance/build_model.py --verify`) and record the result in the README sheet.